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Page No: 1

Date:

Using the bisection method, find a real root of

$$f(x) = x^3 - 4x - 1 = 0 \text{ up to four iterations}$$

$$f(2) = 2^3 - 4 \times 2 - 1 = 8 - 8 - 1 = -1$$

$$f(3) = (3)^3 - 4 \cdot (3) - 1 = 27 - 12 - 1 = 14$$

root lies in interval $[2, 3]$

$$\therefore f(2) \cdot f(3) < 0$$

for iteration 1:-

$$m_1 = \frac{2+3}{2} = 2.5$$

$$f(2.5) = (2.5)^3 - 4 \times (2.5) - 1 = 15.625 - 10 - 1 = 4.625$$

$\therefore$ Root lies in interval $[2, 2.5]$

for iteration 2:-

$$m_2 = \frac{2+2.5}{2} = 2.25$$

$$f(2.25) = 9.595 - 4 \times (2.25) - 1 = 9.595 - 9 - 1 = -0.405$$

$\therefore$ Root lies in interval $[2, 2.25]$

for iteration 4:-

$$m_4 = \frac{2+2.125}{2} = 2.0625$$

$$f(2.0625) = (2.0625)^3 - 4(2.0625) - 1 = -0.4763 < 0$$

$\therefore$ Root lies in interval $[2.0625, 2.125]$

Q2] Apply the bisection method to find the root of

Equation $x \log_{10} x = 1.5$ correct to four decimal place

$$\Rightarrow f(x) = x \log_{10} x - 1.5$$

$$f(3) = 3 \times (0.477) - 1.5 = 1.431 - 1.5 = -0.069$$

$$f(4) = 4 \times (0.602) - 1.5 = 2.408 - 1.5 = 0.908$$

$$f(3) \cdot f(4) < 0$$

Root lies in interval $[3, 4]$

For iteration: 1

$$m_1 = \frac{3+4}{2} = 3.5$$

$$f(3.5) = 3.5 \log(3.5) - 1.5 = 3.5 \times (0.544) - 1.5 = 1.904 - 1.5 = 0.404 > 0$$

Root lies in interval $[3, 3.5]$

For iteration: 2

$$m_2 = \frac{3+3.5}{2} = 3.25$$

$$f(3.25) = 3.25 (0.51188) - 1.5 = 1.6636 - 1.5 = 0.1636 > 0$$

Root lies in interval $[3, 3.25]$

For iteration: 3

$$m_3 = \frac{3+3.25}{2} = 3.125$$

$$f(3.125) = 3.125 (0.4948) - 1.5 = 1.54625 - 1.5 = 0.04625 > 0$$

Root lies in interval $[3, 3.125]$

For iteration 4:

$$m_4 = \frac{3 + 3.125}{2} = 3.0625$$

$$f(3.0625) = (3.0625)(0.48607) - 1.5 = 1.48858 - 1.5 = -0.011420$$

Root lies in interval $[3.0625, 3.125]$

For iteration 5:

$$m_5 = \frac{3.0625 + 3.125}{2} = 3.09375$$

$$f(3.09375) = 3.09375 \log(3.09375) - 1.5 = 3.09375(0.49047) - 1.5 = 1.51786 - 1.5 = 0.0178670$$

Root lies in interval $[3.0625, 3.09375]$

For interval 6:

$$m_6 = \frac{3.0625 + 3.09375}{2} = 3.078125$$

$$f(3.078125) = 3.078125 \log(3.078125) - 1.5 = 1.50273 - 1.5 = 0.00273$$

Root lies in interval $[3.0625, 3.078125]$

For iteration 7:

$$m_7 = \frac{3.0625 + 3.078125}{2} = 3.07031$$

$$f(3.07031) = 3.07031 \log(3.07031) - 1.5 = 1.49578 - 1.5 < 0$$

Root lies in interval $[3.07031, 3.078125]$

For iteration 8:

$$m_8 = \frac{3.07031 + 3.078125}{2} = 3.074215$$

$$f(3.074215) = 3.074215 \log(3.074215) - 1.5 = 1.4992 - 1.5 < 0$$

Root lies in interval $[3.07421, 3.07812]$

for iteration 9

$$m_9 = \frac{3.07421 + 3.07812}{2} = 3.076165$$

$$f(3.07615) = 3.07615 \times 10.4880 - 1.5 = 1.50116 - 1.5 = 0.006170$$

Root lies in interval $[3.07421, 3.076165]$

for iteration 10

$$m_{10} = \frac{3.07421 + 3.07616}{2} = 3.07518$$

$$f(3.07518) = 3.07518 \times 10.4880 - 1.5 = 1.50185 - 1.5 = 0.0018570$$

Root lies in interval $[3.07421, 3.07518]$

for iteration 11

$$m_{11} = \frac{3.07421 + 3.07518}{2} = 3.074695$$

or

$$m_{11} = \frac{3.07421 + 3.07518}{2} \times 10.48779 - 1.5 = -0.0002120$$

Root lies in interval $[3.07469, 3.07518]$

for iteration 12

$$m_{12} = \frac{3.07469 + 3.07518}{2} = 3.074935$$

$$f(3.074935) = 3.074935 \times 10.48779 - 1.5 = 1.500315 = 0.000315$$

Root lies in interval $[3.07469, 3.074935]$

for iteration 13

for iteration 13

$$m_{13} = \frac{3.07469 + 3.074985}{2} = 3.07481$$

$$f(3.0746) = 3.0748 \log(3.07481) - 1.5 < 0$$

Root lies in interval $[3.07481, 3.074935]$

For iteration 14

$$m_{14} = \frac{3.07481 + 3.074935}{2} = 3.074872$$

$\therefore$ Root lies in interval

(Q3)

From a tabulated value of n and y given below, prepare the forward difference table, find the polynomial passing through the points and estimate value of y when $n=11.5$

x 0 2 4 6 8

y 5 25 100 127 175

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	5	$\Delta y_0 = 20$	$\Delta^2 y_0 = 55$	$\Delta^3 y_0 = -103$	$\Delta^4 y_0 = 174$
2	25				
4	100	$\Delta y_1 = 75$	$\Delta^2 y_1 = -48$	$\Delta^3 y_1 = 69$	
6	127	$\Delta y_2 = 27$			
8	175	$\Delta y_3 = 48$	$\Delta^2 y_2 = 21$		

Given

$$u = \frac{n - n_0}{h} = \frac{11.5 - 0}{2} = 5.75$$

$$n=0, \Delta y_0 = 20$$

$$y = y_0 + 4\Delta y_0 + 4(u-1)\Delta^2 y_0 + \frac{4(u-2)(u-3)}{4!}\Delta^3 y_0 + \frac{4(u-2)(u-3)(u-4)}{4!}\Delta^4 y_0$$

$$y(1.5) = 5 + (0.75)(20) + \frac{(0.75)(0.75-1)(50)}{2} + \frac{(0.75)(0.75-1)(0.75-2)(0.75-3)(172)}{6}$$

$$+ \frac{(10.75)(0.75-1)(0.75-2)(0.75-3)(172)}{24}$$

$$y(1.5) = 5 + 15 - 5.1562 - 4.02335 - 3.77929$$

$$y(1.5) = 7.04111$$

For iteration 14

$$m_{14} = \frac{3.07481 + 3.074935}{2} = 3.074872$$

$\therefore$ Root lies in interval

23. Form a tabulated value of n and y given below, prepare the forward difference table. Find the Polynomial passing through the points and estimate value of y when $n=1.5$

x	0	2	4	6	8
y	5	25	100	127	175

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	5	$\Delta y_0 = 20$	$\Delta^2 y_0 = 55$	$\Delta^3 y_0 = -103$	$\Delta^4 y_0 = 172$
2	25				
4	100	$\Delta y_1 = 75$	$\Delta^2 y_1 = -48$	$\Delta^3 y_1 = 69$	
6	127	$\Delta y_2 = 27$			
8	175	$\Delta y_3 = 48$	$\Delta^2 y_2 = 21$		

$$m_{12} = 2.094568$$

Given

$$u \in n_0 - n_0 = \frac{1.8 - 0}{2} = \frac{1.5}{2} = 0.75$$

$$n_0 = 0, D_0 = 20$$

$$y = y_0 + u D_0 + \frac{u^2}{2!} D_0^2 y_0 + \frac{u^3}{3!} D_0^3 y_0 + \frac{u^4}{4!} D_0^4 y_0 + \frac{u^5}{5!} D_0^5 y_0$$

$$y(1.5) = 5 + (0.75)(20) + \frac{(0.75)^2}{2} (155) + \frac{(0.75)^3}{6} (1075) + \frac{(0.75)^4}{24} (1075) + \frac{(0.75)^5}{120} (1075)$$

$$+ 10.75 (0.75)^6$$

using the newton Raphson method, find a root of $n^3 - 2n - 5 = 0$ starting with initial approximation $x_0 = 2$

For up to four iterations

By Newton Raphson method

For iteration 1

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_n)$$

$$f(2) = 2^3 - 2 - 5 = 8 - 9 = -1$$

$$f'(2) = 3(2)^2 - 2 = 10$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f'(x_0)$$

$$x_1 = 2 - \frac{(-1)}{10} = 2 + 0.1 = 2.1$$

$$10$$

For iteration 2

$$f(2.1) = (2.1)^3 - 2 \times (2.1) - 5 = 9.261 - 4.2 - 5$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.061$$

$$f'(x_1)$$

$$f'(2.1) = 3(2.1)^2 - 2 = 13.23 - 2 = 11.23$$

$$x_2 = 2.1 - \frac{0.061}{11.23} = 2.094568$$

$$11.23$$

$$x_2 = 2.094568$$

For iteration 3

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$f(2.094568) = (2.094568)^3 - 2 \times (2.094568)^2 - 5 = 0.000185$$

$$f'(2.094568) = 3(2.094568)^2 - 2 = 11.16164$$

$$x_3 = 2.094568 - \frac{0.000185}{11.16164} \approx 2.094568$$

For iteration 4

$$n_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$x_4 = 2.094568 - \frac{0.000000017}{11.16164} \approx 2.09455148$$

• After four iteration approximate root of eqn 2.09455148

Q5] Apply the newton-Raphson method to find a real root

or $e^x - 3x = 0$ taking initial guess $x_0 = 1$

$$f(x) = e^x - 3x = 0$$

$$x_0 = 1$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x_n)$$

For iteration: 1

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$f'(x_0)$$

$$x_1 = 1 - \frac{f(1)}{f'(1)}$$

$$f'(1)$$

$$f(0) = e^0 - 3 \times (0) = 1$$

$$f'(0) = -3 + 1 = -2$$

$$\therefore n_2 = 0 - \left(\frac{-1}{-2} \right) = 0.5$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \begin{array}{l} f(0.5) = e^{-0.5} - 3 \times (0.5) = 1.6487 - 1.5 \\ = 0.1487 \end{array}$$

$$x_3 = 0.5 - \frac{f(0.5)}{f'(0.5)} \quad \begin{array}{l} f'(0.5) = e^{-0.5} - 3 \approx 1.6487 - 3 \\ = -1.3513 \end{array}$$

$$x_3 = 0.5 + 0.1100 = 0.6100$$

For iteration 4

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$f(0.6100) = e^{-0.6100} - 3(0.6100) = 1.8404 - 1.83 = 0.0104$$

$$f'(0.6100) = e^{-0.6100} - 3 = 1.8404 - 3 = -1.1596$$

$$x_4 = 0.6100 - \left(\frac{0.0104}{-1.1596} \right) = 0.6100 + 0.0090 = 0.6190$$

$\therefore$ the real root of $e^x - 3x = 0$ is approximately
 $[x \approx 0.6190]$

Q6) using the newton-Raphson method, find a root of $\ln(n) + n - 2 = 0$ correct to three decimal places?

$\Rightarrow$ Given: $f(n) = \ln(n) + n - 2 = 0$

$$f(n) = \frac{1}{n} + 1, n_0 = 1$$

M.M.C.O.E.

Page No: 10

Date:

For iteration 1:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 1 - \left[\frac{\ln(1) + 1 - 2}{1 + 1} \right] = 1 - \left[\frac{0 - 1}{2} \right] = 1.5$$

$$x_1 = 1.5$$

For iteration 2:

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = 1.5 - \left[\frac{\ln(1.5) + 1.5 - 2}{\frac{1}{1.5} + 1} \right] = 1.5 - \left[\frac{0.405 + 1.5 - 2}{1.666} \right] = 1.5 - \left[\frac{-0.095}{1.666} \right]$$

$$x_2 = 1.5567$$

For iteration 3:

$$x_3 = 1.5567 - \left[\frac{\ln(1.5567) + 1.5567 - 2}{\frac{1}{1.5567} + 1} \right] = 1.5567 - \left[\frac{0.4425 + 1.5567 - 2}{1.6423} \right]$$

$$x_3 = 1.5567 - (0.00004071) = 1.5571871$$

For iteration 4

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$$

$$= 1.55718 - \left[\frac{\ln(1.55718) + 1.55718 - 2}{1.64221} \right]$$

$$= 1.55718 - \left[\frac{0.4428 + 1.5571 - 2}{1.64221} \right]$$

$$= 1.55718 - (-0.000060893)$$

$$= 1.55718 + 0.000060893$$

$$x_4 = 1.55714$$

Q7] using the newton-Raphson method, find real root of $\cos n - n = 0$ taking approximation $x_0 = 0.5$

$$f(n) = \cos n - n = 0$$

for iteration 1:

$$n_{n+1} = n_n - \frac{f(n_n)}{f'(n_n)}$$

$$n_1 = 0.5 - (\cos(0.5) - 0.5)$$

$$\sin(0.5) + 1$$

$$x_1 = 0.5 + \frac{0.87258 - 0.5}{0.47943 + 1} = 0.5 + \frac{0.39758}{1.47943} \approx 0.75522$$

for iteration 2:

$$x_2 = 0.75522 + \frac{\cos(0.75522) - 0.75522}{\sin(0.75522) + 1}$$

$$\sin(0.75522) + 1$$

$$n_2 = 0.75522 + \frac{0.72829 - 0.75522}{0.685225 + 1} = 0.75522$$

$$0.685225 + 1$$

for iteration 3:

$$x_3 = 0.73914 + \frac{\cos(0.73914) - 0.73914}{\sin(0.73914) + 1} = 0.73914 - 0.0010$$

$$\sin(0.73914) + 1$$

$$1.67366$$

$$[x_3 = 0.73908]$$

The real root of $\cos n - n = 0$ is approximately 0.7391

Q8)

Using Newton's forward interpolation formula find y at $n=2.5$ from following data

x	1	2	3	4	5
y	2	5	10	17	26

→

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
1	2	$\Delta y_0 = 3$	$\Delta^2 y_0 = 2$	
2	5		$\Delta^3 y_0 = 0$	
3	10	$\Delta y_1 = 5$	$\Delta^2 y_1 = 2$	
4	17	$\Delta y_2 = 7$	$\Delta^3 y_1 = 0$	
5	26	$\Delta y_3 = 9$	$\Delta^2 y_2 = 2$	

$$u = \frac{n - n_0}{h} = \frac{2.5 - 1}{1} = 1.5$$

By Newton's forward interpolation

$$y(u) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0$$

$$y(u) = 2 + (1.5 \times 3) + \frac{(1.5 \times 0.5)}{2} \times 2$$

$$y(u) = 2 + 4.5 + 0.75$$

$$[y(2.5) = 7.25]$$

Q9

Q9] using Newton's backward interpolation formula find the value at $n=5.5$ from the following data

x	2	3	4	5	6	7
y	7	13	25	43	67	97

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
2	7					
		6				
3	13		6			
		12		0		
4	25		6		0	
		18		0		0
5	43		6		0	
		24		0		
6	67		6			
		30				
7	97					

$$\rightarrow n_0 = 7, y_0 = 97, u = \frac{n - n_0}{h} = \frac{5.5 - 7}{1} = -1.5$$

$$y = y_0 + u \Delta y + \frac{u(u+1)}{2!} \Delta^2 y$$

$$= 97 + (-1.5) 30 + (-1.5)(-0.5) \times 6$$

$$= 97 - 45 + 2 \cdot 2.5$$

$$\boxed{y = 54.25}$$

Q10] Using Lagrange's interpolation formula find the value of y at $x=2.5$ from following table

x	1	2	3	4
y	1	8	27	64

$$\rightarrow l_0 = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}$$

$$= \frac{(2.5-2)(2.5-3)(2.5-4)}{(1-2)(1-3)(1-4)} = \frac{(0.5)(-0.5)(-1.5)}{(-1)(-2)(-3)} = \frac{0.375}{(-6)} = -0.0625$$

$$l_1 = \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}$$

$$= \frac{(2.5-1)(2.5-3)(2.5-4)(8)}{(2-1)(2-3)(2-4)}$$

$$= \frac{10.125 \times 8}{2} = 4.5$$

$$l_2 = \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}$$

$$= \frac{(2.5-1)(2.5-2)(2.5-4)(27)}{(3-1)(3-2)(3-4)} = \frac{(1.5)(0.5)(-1.5) \times (27)}{2(1)(-1)}$$

$$l_2 = 15.1875$$

$$l_3 = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}$$

$$= \frac{(1.5-1)(1.5-2)(1.5-3)(64)}{(4-1)(4-2)(4-3)} = \frac{(1.5)(0.5)(-0.5)(64)}{6}$$

$$= \frac{-0.375(64)}{6} = -4$$

$$y(1.5) = y_0 l_0(n) + y_1 l_1(n) + y_2 l_2(x) + y_3 l_3(n)$$

$$y(1.5) = -0.0625 + 4.5 + 15.1875 - 4$$

$$y(1.5) = 15.625$$